NOVA

Intelligent Study-Coaching Platform

Git & GitHub Workflow Guide

Symfony 6.4 + JavaFX — 6 Developers

How we collaborate without breaking each other's code

Team & Module Assignment

Each developer owns a dedicated GitHub branch for their module. Nouha is the team lead and is responsible for the initial setup, final merges, and PR reviews. Oussama works directly on main because the Users & Auth module is the foundation that all other modules depend on.

Developer	Module	Branch	Entities
Oussama	Users, Roles & Auth	main (direct)	User, StudentProfile, TutorProfile
Nouha (Team Lead)	Gamification & Rewards	module/gamification	Game, Reward
Wassim	Library Management	module/library	Book, Loan
Acil	Study Sessions & Energy	module/study	PersonalCourse, TutorCourse, Enrollment, StudySession, EnergyState, StudyPlan
Said	Quiz System	module/quiz	Question, Choice
Oumayma	Social Features	module/social	Post, Comment

Why does Oussama push directly to main? Because every other module has Foreign Keys pointing to the User entity. His changes (new fields, role updates, profile changes) affect everyone's code immediately. It is safer for the whole team to always pull from a single source of truth.

Part 1 — Initial Setup (Done Once by Nouha)

Nouha, as team lead, creates all module branches from the current state of main and pushes them to GitHub. This is done once at the beginning so everyone starts from the same codebase.

Run once — create all branches from main
`git checkout main`
`git pull origin main`
`git checkout -b module/gamification`
`git push origin module/gamification`
`git checkout -b module/library`
`git push origin module/library`
`git checkout -b module/study`
`git push origin module/study`
`git checkout -b module/quiz`
`git push origin module/quiz`
`git checkout -b module/social`
`git push origin module/social`
`git checkout main  # return to main after setup`

Part 2 — For Module Developers (Nouha, Wassim, Acil, Said, Oumayma)

Switch to Your Branch

Run once on your machine

```
# Each person switches to THEIR branch only:
git checkout module/gamification   # Nouha
git checkout module/library        # Wassim
git checkout module/study          # Acil
git checkout module/quiz           # Said
git checkout module/social         # Oumayma
```

//whatever name u created your branches as

Daily Workflow — Every Single Day

Always sync with main before starting work. **Oussama's** User entity is the base of the whole project — pulling his latest changes prevents surprises later.

//EACH MEMBER :

Follow these steps every morning before coding

```
# 1. Pull the latest User/Auth changes from main
git checkout main
git pull origin main

# 2. Switch back to your branch and merge main into it
git checkout module/gamification   # your branch
git merge main

# 3. Write your code for the day...

# 4. Stage and commit your work
git add .
git commit -m "example"

# 5. Push to YOUR branch only — never push to main
git push origin module/gamification
```

Commit message format to follow:

- feat(module): short description — for new features
- fix(module): short description — for bug fixes
- refactor(module): short description — for code cleanup
- chore(module): short description — for config/setup changes

Part 3 — For Oussama (Users & Auth — Pushes Directly to Main)

Because the User entity is a foreign key dependency for every other module, Oussama's changes go directly to main. This makes his code the single source of truth for the whole team.

Oussama's daily workflow

```
git checkout main
git pull origin main

# Make your changes to User, StudentProfile, TutorProfile...

git add .
git commit -m "feat(auth): add academicLevel field to StudentProfile"
git push origin main

# ⚠️  IMPORTANT: notify the team after every push to main!
```

> ⚠️ **After every push, send this message to the team (Slack / Discord / WhatsApp):**
> *"🔔 NOVA main updated — please sync your branch before continuing: git checkout main && git pull origin main && git checkout module/YOUR-MODULE && git merge main"*

What to be extra careful about as Oussama:

- Never rename or remove a field that other modules are already using
- If you change the User entity structure, warn the team before pushing
- If you add a new required field, make sure it has a default value so it doesn't break existing data
- Push small, targeted commits — not big batches of changes

Part 4 — What To Do When Oussama Pushes to Main

As soon as you receive the notification from Oussama, stop what you're doing and sync your branch. The longer you wait, the bigger the potential conflict.

Sync your branch with main immediately

```
git checkout main
git pull origin main
```

```
git checkout module/gamification   # your branch
git merge main

# If no conflicts: you're good, keep coding
# If conflicts appear: see conflict resolution below
```

Resolving Merge Conflicts

A conflict happens when both you and Oussama edited the same file. Git will pause the merge and mark the conflicting lines. Don't panic — just follow these steps:

What a conflict looks like inside a file

```
<<<<<<< HEAD
// Your version of the code
=======
// Oussama's version coming from main
>>>>>>> main
```

Steps to resolve:

- Open the conflicting file in your editor
- Read both versions and decide which to keep (or combine them)
- Delete the conflict markers: `<<<<<<<` , `=======` , `>>>>>>>`
- Save the file, then run the commands below

After fixing all conflict files

```
git add .
git commit -m "fix: resolve merge conflict with main (User entity update)"
git push origin module/gamification
```

Part 5 — Cross-Module Dependencies to Watch

Every module in NOVA relies on the User entity. Here is a quick reference of which foreign keys each developer uses from Oussama's module:

Developer	Their Entity	FK to Users module
Nouha (Gamification)	Game	createdBy → User
Wassim (Library)	Loan	userId → User
Acil (Study)	PersonalCourse, TutorCourse, Enrollment, StudySession, EnergyState, StudyPlan	userId / tutorId / studentId → User
Said (Quiz)	Question	createdBy → User
Oumayma (Social)	Post, Comment	userId → User

> ⚠️ **Key rule:**
> *If you are writing code that joins or references the User table and Oussama has recently pushed changes to it, always re-sync your branch before continuing. Otherwise your Doctrine entities or SQL queries may be out of date.*

Part 6 — Final Merge Into Main (Done by Nouha)

When all modules are complete and tested, Nouha merges each branch into main one at a time. Follow this order to minimize conflicts — start with the module that has the fewest dependencies.

Order	Branch	Reason
1st	module/gamification	Reward/Game entities, minimal FK overlap
2nd	module/library	Book/Loan, minimal FK overlap
3rd	module/quiz	Question/Choice depend on TutorCourse — merge after library
4th	module/social	Post/Comment depend on User only
5th	module/study	Most complex — 6 entities, merge last to resolve any remaining conflicts

Team lead merge commands — one branch at a time

```
git checkout main
```

```
git pull origin main

# First module
git merge module/gamification
# Fix conflicts if any, then: git add . && git commit

git merge module/library
# Fix conflicts if any, then: git add . && git commit

git merge module/quiz
git merge module/social
git merge module/study

# Final push
git push origin main
```

Using Pull Requests (Recommended for Final Merge)

Instead of merging locally, open GitHub Pull Requests for better control and code review:

- Each developer opens a PR: their branch → main, when their module is ready
- Nouha (team lead) reviews and merges each PR on GitHub
- GitHub highlights conflicts before merging, preventing surprises
- You get a permanent record of every change — useful for debugging

Part 7 — Team Rules!! IMPORTANT !!

Rule	Who it applies to
Pull from main every morning before coding	Everyone except Oussama
Never push to someone else's branch	Everyone
Oussama announces every push to main via team chat	Oussama
Sync your branch within 30 minutes of Oussama's announcement	Everyone except Oussama
Never rename or delete a User entity field without warning	Oussama
Make small, frequent commits — not huge batches	Everyone

Write descriptive commit messages using the feat/fix/refactor format	Everyone
Test your code locally before pushing	Everyone
Do not merge your own PR — Nouha reviews and merges all PRs	Everyone

Quick Reference Card

🌅 Every morning — before you start

```
git checkout main && git pull origin main
git checkout module/YOUR-MODULE
git merge main
```

💾 Saving your work

```
git add .
git commit -m "feat(module): what you did"
git push origin module/YOUR-MODULE
```

🔔 When Oussama pushes to main — sync NOW

```
git checkout main && git pull origin main
git checkout module/YOUR-MODULE
git merge main
# Fix conflicts if needed, then commit and push
```

🚨 If you have a conflict

```
# 1. Open the file Git flagged
# 2. Remove <<<<<<< ======= >>>>>>> markers
# 3. Keep the correct version of the code
git add .
git commit -m "fix: resolve conflict"
```